

Quiz-1

Subject: Reinforcement learning (CSD366)

Semester: Spring 2025

Date: 12th February 2025

Time: 9.30 AM to 10.15 AM

Full marks: 15 Marks

Answer all the questions. For MCQ questions, the correct response will carry 2 marks whereas wrong answer will attract -1 (negative marking).

1. Which component is essential in Reinforcement learning? [2 Marks]
(a) Agent (b) Environment (c) Rewards (d) All of these (✓).
2. Given that there are m number of *states* and n number of *actions*, the total number of policies will be: [2 Marks]
(a) m^n .
(b) n^m (✓).
(c) $m \times n$.
(d) $\frac{m}{n}$.
3. Reward can be a function of: [2 Marks]
(a) Only state.
(b) Only action.
(c) Both state and action (✓).
(d) None of these.
4. Suppose the reward sequence generated by 5 consecutive states of an MDP is: 4, 2, 3, 1, 9. The discounted reward is (*discount factor is 0.1*): [2 Marks]
(a) 4.2319 (✓)
(b) 3.1297
(c) 4.9314
(d) 2.1879

5. Model the following situation using MDP framework:

[7 Marks]

A robot is roaming inside a shopping mall to collect cans. The robot can take any *one* of the following three actions:

- (a) Actively search for can/garbage.
- (b) Remain stationery and wait for somebody to bring it a can, or
- (c) Go back to the base home and recharge the battery.

Actions are taken depending on the battery power level (High or Low). *The goal of robots is to keep the space as clean as possible.*

Model the situation explicitly mentioning the states, actions, rewards, possible transition probabilities and discount factors. Briefly justify rewards and assigned transition probabilities.

Hint:

States are {High, Low}, actions are {Search, Wait, Recharge}.

Transition probability matrix:

s	s'	a	$p(s' s, a)$	$r(s, a, s')$
high	high	search	α	r_{search}
high	low	search	$1 - \alpha$	r_{search}
low	high	search	$1 - \beta$	-3
low	low	search	β	r_{search}
high	high	wait	1	r_{wait}
high	low	wait	0	r_{wait}
low	high	wait	0	r_{wait}
low	low	wait	1	r_{wait}
low	high	recharge	1	0
low	low	recharge	0	0.

Transition graph:

